

MSc EXAMINATION IN MATHEMATICS AND FINANCE

January 2022

This paper is also taken for the relevant examination for the Associateship.

M5MF3

Stochastic processes (Solutions)

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1. (a) Since both representations are standard we can write

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$$A_j = \cup_{k=0}^N (A_j \cap B_k), \quad B_k = \cup_{j=0}^M (A_j \cap B_k),$$

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for all $1 \leq j \leq M, 1 \leq k \leq N$.

It follows that

$$f(x) = \sum_{j=1}^M \sum_{k=1}^N c_j 1_{A_j \cap B_k}(x), \quad f(x) = \sum_{j=1}^M \sum_{k=1}^N d_j 1_{B_k \cap A_j}(x), \quad (1)$$

For the same reason we can use the finite additivity of μ to get that $\int f d\mu$ is given by

$$\sum_{j=1}^M c_j \mu(A_j) = \sum_{j=1}^M c_j \sum_{k=1}^N \mu(A_j \cap B_k) = \sum_{j=1}^M \sum_{k=1}^N c_j \mu(A_j \cap B_k).$$

On the other hand

$$\sum_{k=1}^N d_k \mu(B_k) = \sum_{k=1}^N d_k \sum_{j=1}^M \mu(A_j \cap B_k) = \sum_{k=1}^N \sum_{j=1}^M d_k \mu(A_j \cap B_k).$$

From (1) we get that $d_k = c_j$ when $A_j \cap B_k \neq \emptyset$. When $A_j \cap B_k = \emptyset$, $\mu(A_j \cap B_k) = 0$, so it follows that

$$\sum_{j=1}^M c_j \mu(A_j) = \sum_{k=1}^N d_k \mu(B_k).$$

- (b) The claim is incorrect. Take $f_n(x) = n 1_{[0,1/n]}(x)$ and note that

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$$\lim_{n \rightarrow \infty} f_n = 0, \quad \text{a.e.}$$

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It follows that

$$\int \lim_{n \rightarrow \infty} f_n d\lambda = 0.$$

where λ is the Lebesgue measure.

On the other hand

$$\lim_{n \rightarrow \infty} \int f_n d\lambda = \lim_{n \rightarrow \infty} 1 = 1.$$

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- (c) Note that

$$|f_n - f|^p \leq (|f_n| + |f|)^p \leq 2^p g^p,$$

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where g is in $L^1(\mu)$. Using reversed Fatou's lemma we get

$$0 = \int \limsup |f_n - f| d\mu \geq \limsup \int |f_n - f| d\mu \geq 0.$$

Therefore

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

2. (a) For any $G \in \mathcal{G}$ we have

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$$E[1_G E[X|\mathcal{G}]] = E[1_G X],$$

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and

$$E[1_G E[\tilde{X}|\mathcal{G}]] = E[1_G \tilde{X}],$$

Hence from monotonicity of the expectation we have

$$E[1_G E[X|\mathcal{G}]] \leq E[1_G E[\tilde{X}|\mathcal{G}]].$$

This holds for any $\mathcal{G}$ -measurable set hence the result follows.

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(b) Clearly $ZE[X|\mathcal{G}]$ is $\mathcal{G}$ -measurable as a product of $\mathcal{G}$ -measurable random variables. Since Z is bounded, there exists a constant $C > 0$ such that

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$$E[|ZE[X|\mathcal{G}]|] \leq CE[|E[X|\mathcal{G}]|] < \infty,$$

where we used the fact that conditional expectation is in L^1 in the second inequality.

Let $G, A \in \mathcal{G}$. Then we have

$$E[1_A 1_B E[X|\mathcal{G}]] = E[1_{A \cap B} E[X|\mathcal{G}]] = E[1_{A \cap B} X].$$

Hence we proved that

$$E[XZ|\mathcal{G}] = ZE[X|\mathcal{G}] \tag{2}$$

for indicator functions. The extension to simple functions is done by linearity of the conditional expectation. When Z is a nonnegative bounded random variable we have an approximating sequence of simple functions $\{Z_n\}$ such that $Z_n \uparrow Z$. Therefore from monotone convergence we get

$$\lim_{n \rightarrow \infty} E[1_A Z_n E[X|\mathcal{G}]] = E[1_A Z E[X|\mathcal{G}]]$$

and by also using part (a) monotone convergence we get

$$\lim_{n \rightarrow \infty} E[1_A E[Z_n X|\mathcal{G}]] = E[1_A E[ZX|\mathcal{G}]].$$

and we get (2). The extension to any bounded Z is done by decomposition to negative Z^- and positive Z^+ parts of Z and using linearity again. Alternatively one can use dominated convergence by showing that the conditions apply.

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(c) Let $Y \in \mathcal{G}$. Using the conditional expectation projection theorem we have

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$$\text{Var}(X - Y) = E[(X - Y)^2] \geq E[(X - E[X|\mathcal{G}])^2]$$

Hence $E[X|\mathcal{G}]$ solves the problem since by conditional Jensen inequality,

$$E[(E[X|\mathcal{G}])^2] \leq E[E[X^2|\mathcal{G}]] = E[X^2] < \infty.$$

and

$$E[E[X|\mathcal{G}]] = E[X] = 0.$$

hence $E[X|\mathcal{G}] \in L_0^2(\mathcal{G})$.

3. (a) Using the independence of X_i^{n+1} , $i = 1, 2, \dots$ we get

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$$\begin{aligned}
 E[\xi^{Z_{n+1}} | \mathcal{F}_n] &= \sum_{k=0}^{\infty} E[\xi^{Z_{n+1}} 1_{\{Z_n=k\}} | \mathcal{F}_n] \\
 &= \sum_{k=1}^{\infty} E[\xi^{\sum_{i=1}^k X_i^{n+1}} 1_{\{Z_n=k\}} | \mathcal{F}_n] \\
 &= \sum_{k=1}^{\infty} 1_{\{Z_n=k\}} E[\xi^{\sum_{i=1}^k X_i^{n+1}} | \mathcal{F}_n] \\
 &= \prod_{k=1}^{\infty} 1_{\{Z_n=k\}} (E[\xi^{X_i^{n+1}}])^k \\
 &= \prod_{k=1}^{\infty} 1_{\{Z_n=k\}} (G(\xi))^k \\
 &= \xi^{Z_n}.
 \end{aligned}$$

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- (b) Since $\{\xi^{Z_n}\}_{n \geq 1}$ is a nonnegative martingale the result follows.
(c) From the martingale property we have

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$$E[\xi^{Z_n}] = E[\xi^{Z_0}] = \xi.$$

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From Fatou's lemma we get

$$E[M_{\infty}] \leq \liminf E[\xi^{Z_n}] \leq \liminf E[\xi^{Z_0}] = \xi.$$

For $\xi \leq 1$ we have $\xi^{Z_n} \leq 1$ for all $n \geq 0$. From dominated convergence it follows that

$$E[M_{\infty}] = \lim_{n \rightarrow \infty} E[\xi^{Z_n}] = \xi.$$

4. (a) Using Ito's Formula we get

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$$dB_t^4 = 4B_t^3 dB_t + 6B_t^2 dt.$$

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Since

$$E \left[\left(\int_0^t 4B_s^3 dB_s \right)^2 \right] = 12E \left[\int_0^t B_s^6 ds \right] = 12E \left[\int_0^t B_s^6 ds \right] < \infty.$$

It follows that

$$E \left[\int_0^t B_s^2 ds \right] = \frac{1}{6} E[B_t^4] = \frac{1}{2} t^2.$$

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(b) Using Ito's formula we get

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$$dR_t^2 = 2R_t dR_t + d\langle R \rangle_t = 2R_t dR_t + \int_0^t g(s)^2 ds.$$

We get

$$dM_t = dR_t^2 - g(t)^2 dt = 2R_t dR_t.$$

Since g is bounded we have

$$\begin{aligned} E \left[\left(2 \int_0^t R_s dR_s \right)^2 \right] &= 4E \left[\left(\int_0^t R_s g(s) dB_s \right)^2 \right] \\ &= 4E \left[\int_0^t R_s^2 g^2(s) ds \right] \\ &\leq CE \left[\int_0^t R_s^2 ds \right]. \end{aligned}$$

Using Jensen inequality and Fubini's theorem we get

$$E \left[\int_0^t R_s^2 ds \right] = \int_0^t E[R_s^2] ds = \int_0^t \int_0^s g(r)^2 dr ds < \infty.$$

Hence M_t is a true martingale.

Review of mark distribution:

Total A marks: 65 of 32 marks

Total B marks: 20 of 20 marks

Total C marks: 10 of 12 marks

Total D marks: 0 of 16 marks

Total marks: 95 of 80 marks

Total Mastery marks: 0 of 20 marks